

21/1/26

SCSB1602 - Network security

Network - collection of devices
 ↳ OSI layer - 7 layer

N/w security protects n/ws & data from attacks & unauthorized access.

Network security

The basic terms of n/w security is confidentiality, integrity, availability.
 ↳ Preventing the loss of data

Confidentiality	Integrity	Availability
Information is safe from accidental or intentional disclosure.	Information is safe from accidental or intentional modification.	Information is available to authorized users when needed.

What's the purpose of CIA

Data is not disclosed eg, encryption	Data is not tampered eg, hashing, digital signature	Data is available eg, backups, redundant systems
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Opposite of CIA

Disclosure	Alteration	Destruction
eg, I send you a msg & no one else knows what that msg is.	eg, I send you a msg & you receive exactly what I sent you (without any modification)	eg, I send you a msg & you are able to receive it.

Intercept (unauthorized)

2m or 12m

OSI security architecture:

Focuses on
security attacks
" mechanisms
" services



Assess effectively the security needs of an organization

1) Security attacks ^{attacker can be}
 → Successful → attacker breaks security
 → Unsuccessful → system stays safe

Refer to the sets of actions that the attackers perform to gain any unauthorized access, cause damage to system/computers, steal data or compromise the computer n/w.

Any action that compromises the security of information owned by an organization

An attacker can launch a cyber attack from any location

The attacker can also be individual or even of groups.

2 types,

- 1) active attack
- 2) passive "

Active

- ⇒ Attacker modifies data
- ⇒ Easy to detect
- ⇒ Involves direct interaction with the system
- ⇒ data may be altered or destroyed
- ⇒ eg, Dos

Passive

- ⇒ Attacker only listens, monitors data
- ⇒ hard to detect
- ⇒ no direct interaction the system
- ⇒ data is not changed
- ⇒ eg, traffic analysis

1) Active attacks

Involve some modification of the data stream or the creation of a false stream

↳ most dangerous bcz sender & receiver may not detect the attack

4 types,

- 1) Masquerade
- 2) Replay
- 3) Modification of messages
- 4) DOS

1) Masquerade

↳ Attacker pretends to be an authorized user

↳ Masquerade take place when one entity

pretends to be a different entity

↳ this is not done by stealing the data packet, it's done by stealing the login details of user some how.

↳ Masquerade attack usually includes

one of the other forms of active attack

↳ eg. masquerade as doctor & fooled everyone.

2) Replay / Playback attack

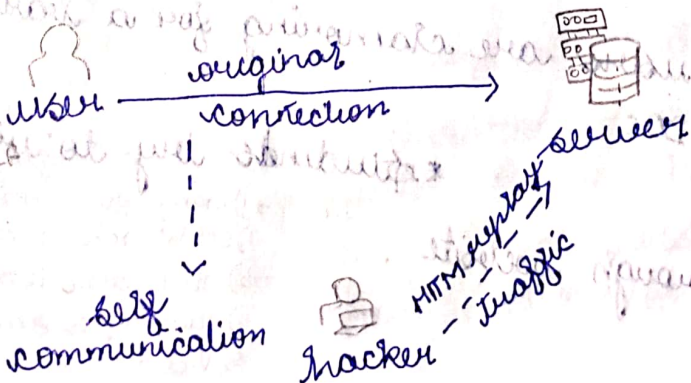
- 1) attacks act as an authorized user
- 2) attacker captures a valid data packet
- 3) later, the attacker sends the same packet again
- 4) server thinks the attacker is a real user

↳ Involves the passive capture of a data unit & its subsequent retransmission to produce an unauthorized effect

↳ Attacker intercepts n/w traffic b/w 2 hosts

↳ The attacker then stores or manipulates the traffic before forwarding it to its intended destination

5] Encryption doesn't stop this attack bcz the packet itself is reused.



3) Modification of message:

↳ some portion of a legitimate message is altered or that messages are delayed or reordered, to produce an unauthorized effect.

↳ eg, Allow Bob to read confidential

file X (original msg)

Allow Bob to read confidential

file Y (hacked msg)

4) Denial of service:

↳ attacks sends too many requests to a server at some time & this is not able to handle.
→ 1) server becomes slow or stop working.
2) requests come from one source (attacker) so, attacker is easy to identify.

↳ A DOS is a form of cyberattack that prevents legitimate users from accessing a computer or n/w.
↳ Another form of service denial is the disruption of an entire n/w, either by disabling the n/w or by overloading it with messages so as to degrade performance.

↳ In a Dos attack, rapid & continuous online requests are sent to a target server to overload the server's bandwidth.

↳ eg,

* An entity may suppress all msg directed to a particular destination.

* Black Friday sales, when thousands of users are clamoring for a bargain, often cause a DOS.

through website.

* Students try to view the result

DDoS → many systems send requests at same time

1) server receive huge traffic & fails

2) Attack source is hard to find.

3) attacking systems are called zombie PCs

4) Victim's system → main victim

users → secondary "

2) Passive attacks → 1) Attacker only watches the data
1) attacker does not change or damage the data
↳ eavesdropping on, or monitoring of, transmissions
3) Sender & receiver do not know about the attack
↳ goals of the opponent is to obtain information that is being transmitted.

2 types:
i) Release of message contents:

↳ A telephone conversation, an electronic mail message, & a transferred file may contain sensitive or confidential informations.

↳ we would like to prevent an opponent from learning the contents of these transmissions.

Eavesdropping → attacker read the actual msg content
Can steal: Pwd, bank details, personal info.
Info. can be used for illegal activities

ii) Traffic analysis:

↳ An opponent is able to observe the pattern of these messages. The opponent could determine the location & identity of communicating hosts & could observe the frequency & length of messages being exchanged.

↳ This information might be useful in guessing the nature of the communication that was taking place.

1) attacker doesn't read the msg
2) attacker observes: amt of data, time & frequency of data transfer
3) from this, attacker can guess information
eg: large data transfer → urgent work
small → less communication

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Active attack

↳ Modification in information takes place

↳ Danger to integrity as well as availability

↳ can be easily detected

↳ the duration of an active attack is short.

↳ the prevention possibility of active attack is high

Passive attack

↳ Modification in information does not take place

↳ Danger to confidentiality

↳ very difficult to detect

↳ the duration of a passive attack is long.

↳ the prevention possibility of passive attack is low.

⇒ Security services used to protect data & systems.

i] Authentication: verify the identity of the users or system.
↳ Authentication is a term that refers to the process of proving that some fact or some document is genuine.

↳ In the case of a single message, such as warning or alarm signal, the function of the authentication service is to assure the recipient that the message is from the source that it claims to be from.

i] Peer entity authentication: verify the identity of the communicating entities.

↳ Provides for the corroboration of the identity of a peer entity in an association.

↳ It is provided for use at the establishment of, or during data transfer phase of a connection.

↳ It attempts to provide confidence that an entity is not performing either a masquerade or an unauthorized replay of a previous connection.

ii] Data origin authentication: verify the source of each individual data unit.

↳ Provides for corroboration of source of a data unit.

↳ It does not provide protection against the duplication or modification of data units.

↳ This type of service supports applications like electronic mail where there are no prior interactions b/w the communicating entities.

2) Access control: that restrict the access to resource based on permissions.

Access control is the ability to limit & control the access to host systems & applications via communications links.

1) Data confidentiality:

Confidentiality is the protection of transmitted data from passive attacks.

2) Data integrity: ensures data is not altered, modified or destroyed during transmission.

↳ Data integrity is the accuracy, completeness, & quality of data as its maintained over time & across formats.

↳ Preserving the integrity of your company's data is a constant process.

3) Non repudiation

Non repudiation prevents either sender or receiver from denying a transmitted msg.

⇒ Security mechanisms used to directly protect data

1. Specific security mechanisms → direct techniques used to provide security service.
2. pervasive security mechanisms

1. specific SM:

* encryption:

Converting plain text [original msg] to cipher text is encryption [encrytion].

The use of mathematical algorithm to transform data into a form that is not readily intelligible.

* Digital signature:

Data appended to, or a cryptographic transformation of, a data unit that allows a recipient of the data unit to prove the source & integrity of the data unit & protect against forgery.

Digital signature

Type of e-signature that (verifies the authenticity of digital document or msg)

Unsigned document $\xrightarrow{\text{S/W}}$ digital signature $\rightarrow$ signed document

↑
line stamped with private key.

2. Persuasive DSM:

* Authentication exchange:

A mechanism intended to ensure the identity of an entity by means of information exchange.

* Traffic padding

The insertion of bits into gaps in a data stream to frustrate traffic analysis attempts

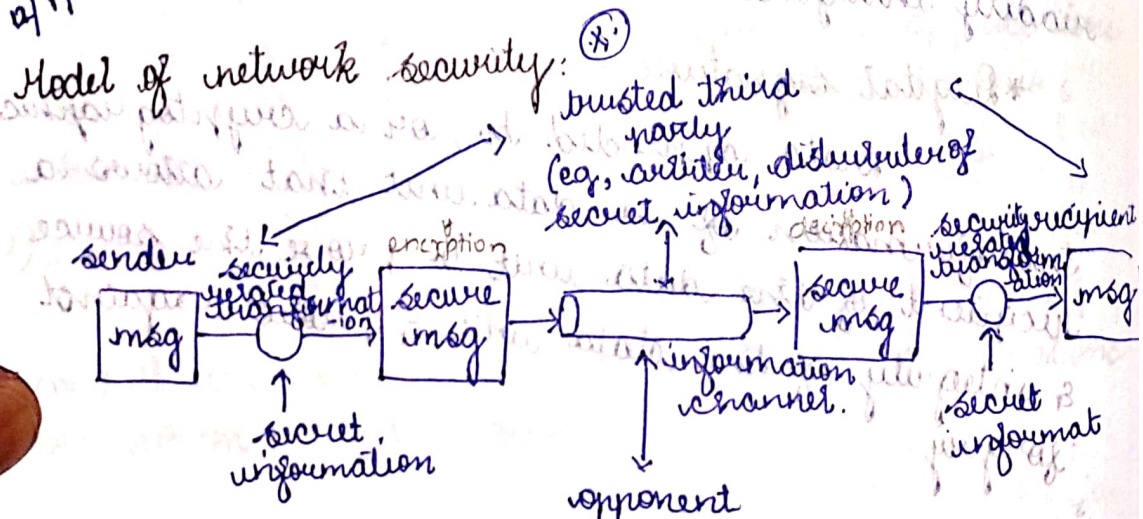
* Routing control

enable selection

* Notarization

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Model of network security:



Msg is to be transferred from one party to another across some sort of internet

2 parties, who are the principal in this transaction must cooperate for the exchange to take place

Logical info. channel is established by defining a route through the internet from source to destination & by the cooperative use of communication protocols (eg, TCP/IP) by 2 principles

Security aspects come into play when it is necessary or desirable to protect the info. transa from an

tasks in designing a particular security service

Design an algorithm for performing the security related transformation. The algorithm should be such that an opponent can't defeat its purpose.

Generate the secret information to be used with the algorithm.

Develop methods for the distribution & sharing of the secret information.

Specify a protocol to be used by the 2 principles that makes use of the security algorithm & secret info. to achieve a particular security service.

Classical encryption techniques

The process of converting from plaintext to ciphertext is known as **encrypting** or **encryption**.

Restoring the plaintext from the ciphertext is **decrypting** or **decryption**.

The many schemes used for encryption constitute the area of study known as **cryptography**.

Techniques used for decrypting a msg without any knowledge of the encrypting details is known as **cryptanalysis**, it is also known as "breaking the code".

The areas of cryptography & cryptanalysis together are called **cryptology**.

A cryptanalyst develops mathematical methods & codes that protect data from computer hackers.

This involves the decryption of a cipher text into plain text in order to transmit a msg over insecure channels.

19/1/26 2 types of encryption,

Symmetric cryptography - In the symmetric cryptography a single key is used for encrypting & decrypting the data.

Encryption

Decryption

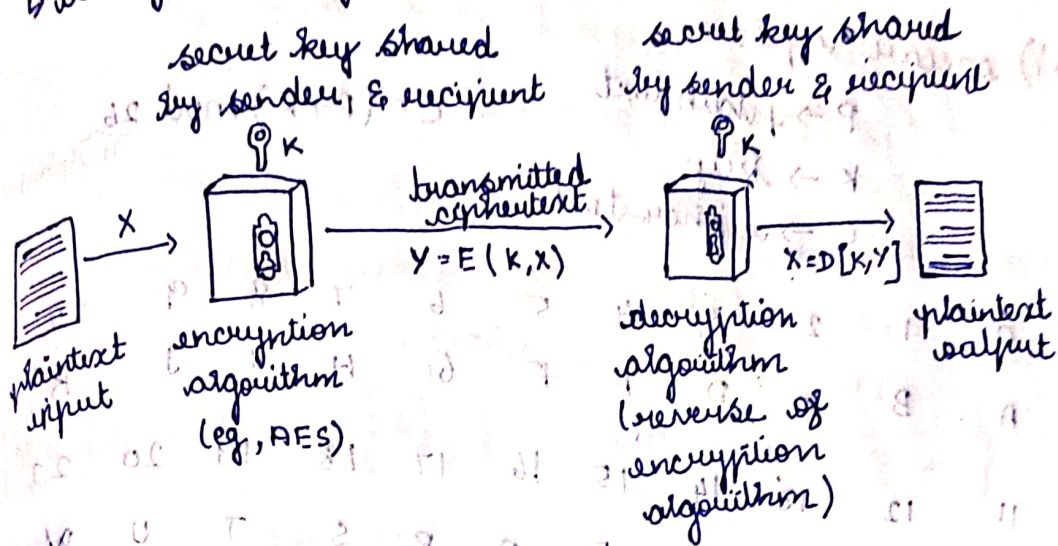
is always done with the same key.

Asymmetric cryptography - In the asymmetric cryptography a pair of key, i.e., public key & private key is used for encryption & decryption.

PKI - Public key Infrastructure.

symmetric cipher model:

- ↳ plain text - superior intelligible msg or data - that fed into
- ↳ encryption algorithm - algorithm as $1/p$
- ↳ secret key
- ↳ ciphertext -
- ↳ decryption algorithm



cryptographic systems are characterized along 3 independent dimensions

1. The types of operations used for transforming plaintext to ciphertext

2. The number of key used. If both sender & receiver use the same key the system is referred to as symmetric, single key, secret key or conventional encryption.

If sender & receiver use different key the system referred as asymmetric, 2-key or public key encryption

3. The way in which plaintext is processed

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2 types of cryptography

1) substitution cipher

↳ Replacing an alphabet with

some other alphabet is called substitution cipher

2) transposition cipher
 ↳ changing the position of the alphabets or symbols in plain text

1) Substitution cipher

↳ i) caesar cipher:

P → plain text

K → key

C → cipher text

$$C = (P + K) \bmod 26$$

↳ alphabets

0	1	2	3	4	5	6	7	8	9	10
A	B	C	D	E	F	G	H	I	J	K
11	12	13	14	15	16	17	18	19	20	21
L	M	N	O	P	Q	R	S	T	U	V
22	23	24	25							
W	X	Y	Z							

Single letter substitution techniques

↳ cipher text is always mentioned in caps. [capital letter]

eg,

o) Plain text: hai, key = 5

h a i
7 0 1

↳ we need take individual letters in sentences separately

Drawback: trying all 26 alphabets

$$C = (P + K) \bmod 26$$

$$= (7 + 5) \bmod 26$$

$$= 12 \bmod 26$$

$$= 12$$

C [cipher] = M

Plaintext → helloworld

keywork → nlw

1) Brute force attack

2) Monoalphabetic substitution cipher

3) Polyalphabetic cipher

↳ lesser than mod value put the same value

1) vigenere cipher

2) veeenam cipher

$$C = (0 + 5) \bmod 26$$

$$= 5 \bmod 26$$

$$= 5$$

$$\boxed{C = F}$$

$$C = (8 + 5) \bmod 26$$

$$= 13 \bmod 26$$

$$= 13$$

$$\boxed{C = N}$$

$\therefore C = MFN$ where $P = \text{hai}$

$$\boxed{P = (C - K) \bmod 26}$$

Q) $C = MFN$, $\text{key} = 5$

$$C = M F N$$

$$12 \ 5 \ 13$$

$$P = (12 - 5) \bmod 26$$

$$= 7 \bmod 26$$

$$= 7$$

$$\boxed{P = H}$$

$$P = (13 - 5) \bmod 26$$

$$= 8 \bmod 26$$

$$= 8$$

$$\boxed{P = i}$$

$\therefore P = \text{hai}$ where $C = MFN$

$$P = (5 - 5) \bmod 26$$

$$= 0 \bmod 26$$

$$= 0$$

$$\boxed{P = a}$$

Q) $P \rightarrow \text{sathyabama}$

$K \rightarrow 6$

$C = ?$

Encrypt & Decrypt

$$C = (P + K) \bmod 26 \rightarrow \text{for encryption}$$

$P \rightarrow \text{s a t h y a b a m a}$

18 0 19 7 24 0 1 0 12 0

$$C = (18 + 6) \bmod 26$$

$$= 24 \bmod 26$$

$$= 24$$

$$C = Y$$

$$C = (19 + 6) \bmod 26$$

$$= 25 \bmod 26$$

$$= 25$$

$$C = Z$$

$$C = (24 + 6) \bmod 26$$

$$= 30 \bmod 26$$

$$\begin{array}{r} 1 \\ 26 \overline{) 30} \\ \underline{26} \\ 4 \end{array}$$

$$= 4$$

$$C = E$$

$$C = (12 + 6) \bmod 26$$

$$= 18 \bmod 26$$

$$= 18$$

$$C = S$$

$$C = (0 + 6) \bmod 26$$

$$= 6 \bmod 26$$

$$= 6$$

$$C = G$$

$$C = (7 + 6) \bmod 26$$

$$= 13 \bmod 26$$

$$= 13$$

$$C = N$$

$$C = (1 + 6) \bmod 26$$

$$= 7 \bmod 26$$

$$= 7$$

$$C = H$$

$\therefore C = YGZNEGHGSG$ where $P = \text{sathyabama}$

$$P = (C - K) \bmod 26 \rightarrow \text{for decryption}$$

$$C = Y \quad G \quad Z \quad N \quad E \quad G \quad H \quad G \quad S \quad G$$

$$24 \quad 6 \quad 25 \quad 13 \quad 4 \quad 6 \quad 7 \quad 6 \quad 18 \quad 6$$

$$P = (24 - 6) \bmod 26$$

$$= 18 \bmod 26$$

$$= 18$$

$$P = (6 - 6) \bmod 26$$

$$= 0 \bmod 26$$

$$= 0$$

$$P = S$$

$$P = a$$

$$P = (25 - 6) \bmod 26$$

$$= 19 \bmod 26$$

$$= 19$$

$$P = (13 - 6) \bmod 26$$

$$= 7 \bmod 26$$

$$= 7$$

$$P = t$$

$$P = h$$

$$P = (4 - 6) \bmod 26$$

$$= -2 \bmod 26$$

$$= +24 \Rightarrow -2 + 26$$

$$P = (7 - 6) \bmod 26$$

$$= 1 \bmod 26$$

$$= 1$$

$$P = Y \Rightarrow 24$$

$$P = b$$

$$P = (18 - 6) \bmod 26$$

$$= 12 \bmod 26$$

$$= 12$$

$$P = m$$

P = Sathyaabama where C = Y G Z N E G H G S G

ii) Playfair cipher

↳ Multi letter substitution technique

↳ It is multi letter encryption

technique

↳ In this eg plain text is Sallan,

keyword is monarchy & we need to find cipher & then the decryption

Step 1:

Draw a 5×5 matrix

Step 2:

Start filling the matrix with the given keyword

☞ In this eg as the keyword is monarchy the grid is filled with monarchy

Step 3:

Fill the remaining grid starting from alphabet A provided if the alphabet are in the keyword leave the alphabet & try soon (no duplication of alphabets in the matrix)

Step 4:

I/J is in the same cell as the frequency of forming a word with the combination of 2 letters (researcher proved)

Step 5:

Split the plain text 2 letters per pair with the condition that no duplication of letters in same pair

step 6: if there is any duplication in same pair insert dummy value n then split
 n is the plain text use dummy value y

step 7: if the pair of letters in the plain text lies in same row the cipher text will be alphabet will be the right of the alphabet

step 8: if the pair of letters lies in same column the cipher text will be the alphabet down alphabet

step 9: if the pairs lies in the intersection neither row nor column cipher will be the row wise intersection of alphabet.

eg, [encryption]

P → balloon

Keyword → monarchy

5x5
 some letter in pair not allowed?

ba ll oo n

insert n,

ba lx lo on

ls B su PH NA

column E → down

row → right side word

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

→ Pairs are present in same column it see the down word next to it

→ alphabets not repeated in keyword

⇒ if suppose TZ are pair we take T ⇒ Z But for Z ⇒ R we take For ST ⇒ S ⇒ T, T ⇒ L

never consider the space, symbol in plain text.

Q) Plaintext $\rightarrow$ hello world

keyword $\rightarrow$ Xavier

Encryption:
5x5

↑ keyword

X	A	V	I	E
R	B	C	D	F
G	H	J	K	L
M	N	O	P	S
T	U	W	Y	Z

$\rightarrow$ If I/J is present keyword u don't anyother pair so u can omit a alphabet or instead of write I/J we omit a & write I/J is different grid

$\Rightarrow$ here i omit a instead of writing I/J

$\Rightarrow$ If n is present in Plaintext insert -end of n.

He ll ow or ld
inserting n:
He ll ow or ld
He ll ow or ld
LA GE JS VW FG RI

LA GE JS VW FG RI

Find H value by intersection of H & E So to find H we need take row of H & column of E
for E $\rightarrow$ row of E & column of H
first $\rightarrow$ rowwise &

In playfair cipher

Encryption $\Rightarrow$ column $\rightarrow$ down
row $\rightarrow$ right

Decryption $\Rightarrow$ column $\rightarrow$ up
row $\rightarrow$ left

decryption

He ll ow or ld
 $\downarrow$ LA GE JS VW FG RI
 $\downarrow$ He ll ow or ld

2	4	4	0	11
0	2	4	4	0
2	2	0	7	7
7	2	0	9	1
2	2	4	4	0

Drawback in Vigenere cipher

- i) Brute force attack
- ii) Monoalphabetic substitution cipher
- iii) Polyalphabetic cipher

1) Vigenere cipher

↳ condition: length of plaintext & keyword should be same.

eg,

Plaintext → helloworld

Keyword → network

$$\begin{matrix} P \rightarrow & h & e & l & l & o & w & o & r & l & d \\ & 7 & 4 & 11 & 11 & 14 & 22 & 14 & 17 & 11 & 13 \end{matrix}$$

$$\begin{matrix} K \rightarrow & n & e & t & w & o & r & k & n & e & t \end{matrix}$$

(+)
$$\begin{matrix} 7 & 4 & 11 & 11 & 14 & 22 & 14 & 17 & 11 & 13 \\ + & 13 & 4 & 19 & 22 & 14 & 17 & 10 & 13 & 4 & 19 \end{matrix}$$

$$\begin{matrix} 20 & 8 & 30 & 33 & 28 & 39 & 24 & 30 & 15 & 22 \end{matrix}$$

$$\begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \end{matrix}$$

$$\begin{matrix} (20 \times 26) + (8 \times 26) + (30 \times 26) + (33 \times 26) + (28 \times 26) + (39 \times 26) + (24 \times 26) + (30 \times 26) + (15 \times 26) + (22 \times 26) \\ (U \times 26) + (I \times 26) + (E \times 26) + (H \times 26) + (C \times 26) + (N \times 26) + (Y \times 26) + (E \times 26) + (P \times 26) + (W \times 26) \end{matrix}$$

$$C = (P + K) \bmod 26$$

$$C = 33 \bmod 26$$

$$C = 28 \bmod 26$$

$$\begin{matrix} 1 \\ 26 \overline{) 30} \\ 26 \\ \hline 4 \end{matrix}$$

$$= 30 \bmod 26$$

$$= 7 \Rightarrow H$$

$$= 2 \Rightarrow C$$

$$= 4 \Rightarrow E$$

$$C = 39 \bmod 26$$

$$= 13 \Rightarrow N$$

$$C = 24 \bmod 26$$

$$= 24 \Rightarrow Y$$

$$C = 20 \bmod 26$$

$$= 20 \Rightarrow U$$

2) Vernam cipher

↳ applicable only for binary no.

eg, Plaintext 0 1 0 1 1 0 1 0 0

Keyword (+) 1 0 1 0 0 0 1 1 1

$$\begin{matrix} 0 & 1 & 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ + & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 1 & 1 & 0 & 0 & 1 & 1 \end{matrix}$$

Same → 0

diff → 1

$$P \oplus K$$

XOR.

Hill cipher

- ↳ Lester hill
- ↳ multi letter encryption techniques
- ↳ based on the size of matrix (eg, 2×2 , 3×3)
- the plain text is split.

↳ formula, $C = K P \text{ mod } 26$

$$\begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix} = \begin{pmatrix} K_{11} & K_{12} & K_{13} \\ K_{21} & K_{22} & K_{23} \\ K_{31} & K_{32} & K_{33} \end{pmatrix} \begin{pmatrix} P_1 \\ P_2 \\ P_3 \end{pmatrix} \text{ mod } 26$$

$$\begin{array}{r} 52 \\ 79 \ 4 \\ \hline 26 \times 7 \\ 182 \\ \hline 13 \end{array}$$

eg, keyword $\begin{pmatrix} 17 & 17 & 4 \\ 2 & 18 & 10 \\ 3 & 5 & 19 \end{pmatrix}$ plaintext is paymorfemofney

$$\begin{array}{r} 26 \mid 351 \\ 26 \downarrow \\ 91 \\ 79 \\ \hline 12 \end{array}$$

pay,

$$\begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix} = \begin{pmatrix} 17 & 17 & 4 \\ 2 & 18 & 10 \\ 3 & 5 & 19 \end{pmatrix} \begin{pmatrix} 15 \\ 6 \\ 24 \end{pmatrix} \text{ mod } 26$$

$$\begin{pmatrix} (17 \times 15) + (17 \times 6) + (4 \times 24) \\ (2 \times 15) + (18 \times 6) + (10 \times 24) \\ (3 \times 15) + (5 \times 6) + (19 \times 24) \end{pmatrix} \text{ mod } 26$$

$$= \begin{pmatrix} 351 \\ 210 \\ 501 \end{pmatrix} \text{ mod } 26$$

$$= \begin{pmatrix} 13 \\ 10 \\ 7 \end{pmatrix} \begin{matrix} N \\ K \\ 17 \end{matrix}$$

$$\begin{array}{r} 001011010 \\ 111000101 \\ \hline 110011111 \end{array}$$

$$\text{mod } 9 \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 17 & 17 & 4 \\ 2 & 18 & 10 \\ 3 & 5 & 19 \end{pmatrix} \begin{pmatrix} 12 \\ 14 \\ 17 \end{pmatrix} \text{mod } 26$$

$$= \begin{pmatrix} (17 \times 12) + (17 \times 14) + (4 \times 17) \\ (2 \times 12) + (18 \times 14) + (10 \times 17) \\ (3 \times 12) + (5 \times 14) + (19 \times 17) \end{pmatrix} \text{mod } 26$$

$$\begin{array}{r} 2 \\ 5 \times 14 \\ \hline 70 \end{array}$$

$$\begin{array}{r} 2 \\ 3 \times 17 \\ \hline 51 \end{array}$$

$$\begin{array}{r} 17 \times 12 \\ \hline 204 \end{array}$$

$$\begin{array}{r} 17 \times 14 \\ \hline 238 \end{array}$$

$$\begin{array}{r} 4 \times 17 \\ \hline 68 \end{array}$$

$$\begin{pmatrix} 204 + 238 + 68 \\ 24 + 252 + 170 \\ 36 + 70 + 323 \end{pmatrix} \text{mod } 26$$

$$= \begin{pmatrix} 510 \\ 446 \\ 429 \end{pmatrix} \text{mod } 26$$

$$= \begin{pmatrix} 16 \\ 14 \\ 13 \end{pmatrix}$$

$$\begin{pmatrix} 16 \\ 14 \\ 13 \end{pmatrix} \text{ E N }$$

$$\begin{array}{r} 12 \\ 204 \\ 238 \\ \hline 68 \end{array}$$

$$\begin{array}{r} 3 \\ 26 \times 6 \\ \hline 156 \end{array}$$

$$\begin{array}{r} 26 \overline{) 429} \\ 26 \downarrow \\ \hline 169 \end{array}$$

$$\begin{array}{r} 446 \\ 26 \downarrow \\ \hline 169 \end{array}$$

$$\begin{array}{r} 156 \\ 26 \downarrow \\ \hline 13 \end{array}$$

emo,

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 17 & 17 & 4 \\ 2 & 18 & 10 \\ 3 & 5 & 19 \end{pmatrix} \begin{pmatrix} 4 \\ 12 \\ 14 \end{pmatrix} \text{mod } 26$$

$$\begin{array}{r} 238 \\ 204 \\ \hline 56 \end{array}$$

$$\begin{array}{r} 16 \quad 266 \\ 140 \quad 60 \\ 8 \quad 12 \\ \hline 364 \quad 338 \end{array}$$

$$\begin{array}{r} 26 \overline{) 498} \\ 26 \downarrow \\ \hline 238 \end{array}$$

$$= \begin{pmatrix} (17 \times 4) + (17 \times 12) + (4 \times 14) \\ (2 \times 4) + (18 \times 12) + (10 \times 14) \\ (3 \times 4) + (5 \times 12) + (19 \times 14) \end{pmatrix} \text{mod } 26$$

$$\begin{array}{r} 4 \times 14 \\ \hline 56 \end{array}$$

$$\begin{array}{r} 18 \times 12 \\ \hline 216 \end{array}$$

$$= \begin{pmatrix} 238 + 204 + 56 \\ 8 + 216 + 140 \\ 12 + 60 + 266 \end{pmatrix} \text{mod } 26$$

$$\begin{array}{r} 26 \overline{) 338} \\ 26 \downarrow \\ \hline 78 \end{array}$$

$$\begin{array}{r} 19 \times 14 \\ \hline 266 \end{array}$$

$$= \begin{pmatrix} 364 \\ 338 \end{pmatrix} \text{mod } 26$$

key,

$$\begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix} = \begin{pmatrix} 17 & 17 & 4 \\ 2 & 18 & 10 \\ 3 & 5 & 19 \end{pmatrix} \begin{pmatrix} 13 \\ 4 \\ 24 \end{pmatrix} \begin{matrix} n \\ e \\ y \end{matrix}$$

$$= \begin{pmatrix} (17 \times 13) + (17 \times 4) + (4 \times 24) \\ (2 \times 13) + (18 \times 4) + (10 \times 24) \\ (3 \times 13) + (5 \times 4) + (19 \times 24) \end{pmatrix}$$

$$= \begin{pmatrix} 221 + 68 + 96 \\ 26 + 72 + 240 \\ 39 + 20 + 456 \end{pmatrix}$$

$$= \begin{pmatrix} 385 \\ 338 \\ 515 \end{pmatrix}$$

$$= \begin{pmatrix} 21 \\ 0 \\ 21 \end{pmatrix} \begin{matrix} V \\ A \\ V \end{matrix}$$

$$\begin{array}{r} 11 \\ 221 \\ 68 \\ 96 \\ \hline 385 \\ 240 \\ 72 \\ 26 \\ \hline 338 \end{array} \quad \begin{array}{r} 11 \\ 456 \\ 39 \\ 20 \\ \hline 515 \end{array}$$

$$\begin{array}{r} 3 \\ 19 \times 24 \\ 216 \\ 24 \\ \hline 456 \\ 32 \\ 4 \\ \hline 72 \\ 17 \times 13 \\ 221 \\ 13 \\ \hline 221 \\ 2 \\ \hline 17 \times 4 \\ 68 \\ 4 \\ \hline 3 \times 13 \\ 39 \\ 68 \\ 24 \times 4 \\ 96 \\ 3 \\ 16 \\ \hline 19 \times 24 \\ 216 \\ 24 \\ \hline 456 \\ 32 \\ 4 \\ \hline 72 \\ 17 \times 13 \\ 221 \\ 13 \\ \hline 221 \\ 2 \\ \hline 17 \times 4 \\ 68 \\ 4 \\ \hline 3 \times 13 \\ 39 \\ 68 \\ 24 \times 4 \\ 96 \end{array}$$

N K H Q E N Q A A V A V → encrypted.

Decryption → $P = K^{-1} C \text{ mod } 26 \Rightarrow K^{-1} = \frac{1}{\det(K)} \text{adj}(K)$

$$\det(K) = 17 \begin{vmatrix} 18 & 19 \\ 2 & 5 \end{vmatrix} - 17 \begin{vmatrix} 2 & 18 \\ 3 & 19 \end{vmatrix} + 4 \begin{vmatrix} 2 & 18 \\ 3 & 19 \end{vmatrix}$$

$$= 17(292) - 17(8) + 4(-44)$$

$$= 4964 - 136 - 176$$

$$= 4652$$

$$K^{-1} = \begin{pmatrix} 17 & 2 & 3 \\ 17 & 18 & 5 \\ 4 & 10 & 19 \end{pmatrix}$$

$$\text{adj}(K) = \begin{bmatrix} 18 \times 19 - 10 \times 52 & 17 \times 19 - 5 \times 4 & 17 \times 10 - 18 \times 4 \\ 2 \times 19 - 3 \times 10 & 17 \times 19 - 4 \times 3 & 17 \times 10 - 2 \times 4 \\ 2 \times 5 - 18 \times 3 & 17 \times 5 - 17 \times 3 & 17 \times 18 - 17 \times 2 \end{bmatrix}$$

$$= \begin{bmatrix} (342 - 50) & (323 - 20) & (170 - 72) \\ (38 - 30) & (323 - 12) & (170 - 8) \\ (10 - 54) & (85 - 51) & (306 - 34) \end{bmatrix} = \begin{bmatrix} 292 & -303 & 98 \\ -8 & 311 & -162 \\ -44 & -34 & 272 \end{bmatrix}$$

$$\begin{pmatrix} P_1 \\ P_2 \\ P_3 \end{pmatrix} = \frac{1}{441} \begin{bmatrix} 292 & -303 & 98 \\ -8 & 311 & -162 \\ -44 & -34 & 272 \end{bmatrix} \begin{bmatrix} 13 \\ 10 \\ 7 \end{bmatrix} \pmod{26}$$

$$= \frac{1}{441} \begin{bmatrix} (292 \times 13) + (-303 \times 10) + (98 \times 7) \\ (-8 \times 13) + (311 \times 10) + (-162 \times 7) \\ (-44 \times 13) + (-34 \times 10) + (272 \times 7) \end{bmatrix} \pmod{26}$$

$$= \frac{1}{441} \begin{bmatrix} 3796 + (-3030) + 686 \\ -104 + 3110 + (-1134) \\ -572 + (-340) + 1904 \end{bmatrix} \pmod{26} = \frac{1}{441} \begin{bmatrix} 1452 \\ 1872 \\ 992 \end{bmatrix} \pmod{26}$$

$$\leq \frac{1}{441} \begin{bmatrix} 1452 \\ 1872 \\ 992 \end{bmatrix} \pmod{26} = \begin{bmatrix} 4 \\ 0 \\ 22 \end{bmatrix} \begin{matrix} E \\ A \\ W \end{matrix}$$

QEN,

$$\begin{pmatrix} P_1 \\ P_2 \\ P_3 \end{pmatrix} = \frac{1}{441} \begin{bmatrix} 292 & -303 & 98 \\ -8 & 311 & -162 \\ -44 & -34 & 272 \end{bmatrix} \begin{bmatrix} 16 \\ 4 \\ 13 \end{bmatrix} \pmod{26}$$

$$= \frac{1}{441} \begin{bmatrix} (292 \times 16) + (-303 \times 4) + (98 \times 13) \\ (-8 \times 16) + (311 \times 4) + (-162 \times 13) \\ (-44 \times 16) + (-34 \times 4) + (272 \times 13) \end{bmatrix} \pmod{26}$$

$$= \frac{1}{441} \begin{bmatrix} 4734 \\ -990 \\ 2696 \end{bmatrix} \pmod{26} = 25 \begin{bmatrix} 4734 \\ -990 \\ 2696 \end{bmatrix} \pmod{26} = \begin{bmatrix} 118,350 \\ -24,750 \\ 67,400 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 24 \\ 24 \\ 24 \end{bmatrix}$$

17x10

$$\begin{array}{r} 38 \\ 26 \overline{) 1872} \\ \underline{1820} \\ 52 \\ \underline{52} \\ 0 \end{array}$$

$$\begin{array}{r} 55 \\ 26 \overline{) 1452} \\ \underline{1300} \\ 152 \\ \underline{130} \\ 22 \end{array}$$

$$\begin{array}{r} 24 \\ 26 \overline{) 67400} \\ \underline{6240} \\ 5000 \\ \underline{4940} \\ 600 \\ \underline{578} \\ 22 \end{array}$$

Steganography:

Steganography is the technique of hiding secret information inside another normal file (like an image, audio, video or text) so that no one knows a message exists.

Unlike cryptography (which hides the content), steganography hides the existence of the message.

Types of steganography:

Text steganography	
Image	"
Audio	"
Video	"
network	"

Text steganography

↳ secret message is hidden inside text

↳ Methods:

Extra spaces
capital letters

font changes

↳ eg, using spaces between words

to represent binary data

2. Image steganography

↳ Data is hidden inside image pixels

↳ Usually modifies the least significant

bit (LSB) of pixel values

↳ eg, hiding text inside a JPG or PNG

image

3. Audio steganography

↳ secret data hidden inside audio files

↳ Methods:

Echo hiding

LSB modification

↳ eg, hiding data in a song file

4. Video steganography

↳ combines image & audio steganography

↳ data hidden in video frames or

sound

↳ eg, secret msg inside an MP4

file

5. Network steganography:

- ↳ Data hidden in network packets
- ↳ uses protocol fields or timing
- ↳ eg, hiding data in TCP/IP headers

Advantages:

1. High secrecy - no one suspects hidden data
2. Hard to detect - looks like a normal file
3. can be combined with cryptography - extra security
4. Maintains file appearance - no visible change
5. Useful for covert communication

Application:

1. Secure communication
 - ↳ military & intelligence agencies
2. Data protection
 - ↳ protect confidential info.
3. Digital watermarking
 - ↳ copyright protection for images & videos.
4. Authentication
 - ↳ verify originality of digital files.
5. Cybersecurity
 - ↳ secure data transfer
 - ↳ malware analysis (also used by hacker)
6. Journalism
 - ↳ safe communication in sensitive regions

Asymmetric key cryptography

same key is used by the sender (for encryption) & receiver (decryption)

Always encryption done by the sender's public key

Always decryption done by the receiver's private key to ensure the integrity & authentication

Asymmetric key

Alice



Plaintext

Encryption

Cipher Text

Decryption

Plaintext



Bob

To the Public

Bob's Public Key

Bob's Private Key

Asymmetric algorithm

RSA

Digital Signature

Categories

Symmetric key cryptography

Asymmetric key cryptography

key used for encryption / decryption

same key is used for encryption & decryption

one key is used for encryption, another different key is used for decryption

key process

$$K_e = K_d$$

$$K_e \neq K_d$$

Speed of encryption / decryption

very fast

Slower

size of resulting encrypted text

Usually same as or less than the original clear text size

more than the original clear-text

Key agreement/exchange

a public problem

no problem at all

Usage

Mainly used for encryption & decryption, can't be used for digital signatures

can be used for encryption & decryption as well as for digital signature

efficiency in usage

Symmetric key cryptography is often used for long msgs

Asymmetric key cryptography are more efficient for short msgs

Stream - bit by bit or byte by byte.

block - take block 64 bits

block vs stream cipher

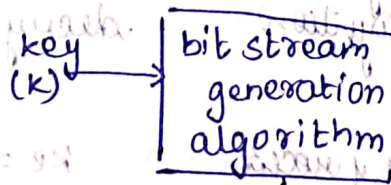
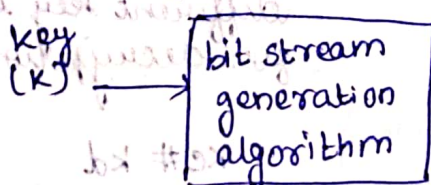
↳ block cipher messages in blocks, each of which is the encrypted.

↳ like a substitution on very big characters

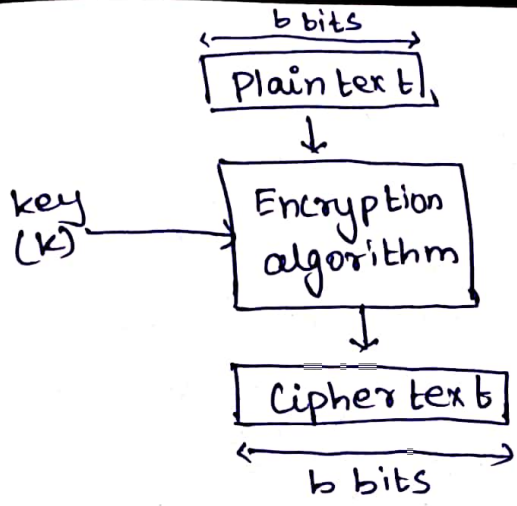
64 bits or more

↳ stream cipher process messages a bit or byte at a time when encrypting

↳ many current ciphers are block ciphers



(a) Stream cipher using Algorithm bit stream cipher



(b) block cipher.